

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z3,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z3 ZL: $\frac{Z_1 Z_3 Z_L g_m}{Z_1 Z_3 g_m + Z_1 Z_L g_m + Z_3 + Z_L}$

$$H(z) = \frac{Z_1 Z_3 Z_L g_m}{Z_1 Z_3 g_m + Z_1 Z_L g_m + Z_3 + Z_L}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_3 g_m}{C_1 C_L R_3 s^2 + g_m + s (C_1 + C_L R_3 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_3} \sqrt{g_m}}{C_1 + C_L R_3 g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_3}}$
bandwidth: $\frac{C_1 + C_L R_3 g_m}{C_1 C_L R_3}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_3 R_L g_m}{C_1 C_L R_3 R_L s^2 + R_3 g_m + R_L g_m + s (C_1 R_3 + C_1 R_L + C_L R_3 R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L} \sqrt{g_m} \sqrt{R_3 + R_L}}{C_1 R_3 + C_1 R_L + C_L R_3 R_L g_m}$
wo: $\frac{\sqrt{R_3 g_m + R_L g_m}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}}$
bandwidth: $\frac{\sqrt{R_3 g_m + R_L g_m} (C_1 R_3 + C_1 R_L + C_L R_3 R_L g_m)}{C_1 C_L R_3 R_L \sqrt{g_m} \sqrt{R_3 + R_L}}$
K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_L} \sqrt{g_m}}{C_1 + C_3 R_L g_m}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 + C_3 R_L g_m}{C_1 C_3 R_L}$
 K-LP: R_L
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

$$H(s) = \frac{R_L g_m}{C_1 C_3 R_L s^2 + g_m + s(C_1 + C_3 R_L g_m)}$$

7.4 LP-4 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_L} \sqrt{g_m} \sqrt{C_3 + C_L}}{C_1 + C_3 R_L g_m + C_L R_L g_m}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_3 R_L + C_1 C_L R_L}}$
 bandwidth: $\frac{C_1 + C_3 R_L g_m + C_L R_L g_m}{\sqrt{C_1} \sqrt{R_L} \sqrt{C_3 + C_L} \sqrt{C_1 C_3 R_L + C_1 C_L R_L}}$
 K-LP: R_L
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

$$H(s) = \frac{R_L g_m}{g_m + s^2(C_1 C_3 R_L + C_1 C_L R_L) + s(C_1 + C_3 R_L g_m + C_L R_L g_m)}$$

7.5 LP-5 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_L} \sqrt{g_m} \sqrt{R_3 + R_L}}{C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m}$
 wo: $\frac{\sqrt{R_3 g_m + R_L g_m}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_L}}$
 bandwidth: $\frac{\sqrt{R_3 g_m + R_L g_m} (C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m)}{C_1 C_3 R_3 R_L \sqrt{g_m} \sqrt{R_3 + R_L}}$
 K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

$$H(s) = \frac{R_3 R_L g_m}{C_1 C_3 R_3 R_L s^2 + R_3 g_m + R_L g_m + s(C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m)}$$

7.6 LP-6 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{1}{C_L s} \right)$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_3} \sqrt{g_m} \sqrt{C_3 + C_L}}{C_1 + C_3 R_3 g_m + C_L R_3 g_m}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_3 R_3 + C_1 C_L R_3}}$
 bandwidth: $\frac{C_1 + C_3 R_3 g_m + C_L R_3 g_m}{\sqrt{C_1} \sqrt{R_3} \sqrt{C_3 + C_L} \sqrt{C_1 C_3 R_3 + C_1 C_L R_3}}$

$$H(s) = \frac{R_3 g_m}{g_m + s^2(C_1 C_3 R_3 + C_1 C_L R_3) + s(C_1 + C_3 R_3 g_m + C_L R_3 g_m)}$$

K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.7 LP-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_3 R_L g_m}{R_3 g_m + R_L g_m + s^2 (C_1 C_3 R_3 R_L + C_1 C_L R_3 R_L) + s (C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m + C_L R_3 R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_3} \sqrt{R_L} \sqrt{g_m} \sqrt{C_3 + C_L} \sqrt{R_3 + R_L}}{C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m + C_L R_3 R_L g_m}$
wo: $\frac{\sqrt{R_3 g_m + R_L g_m}}{\sqrt{C_1 C_3 R_3 R_L + C_1 C_L R_3 R_L}}$
bandwidth: $\frac{\sqrt{R_3 g_m + R_L g_m} (C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m + C_L R_3 R_L g_m)}{\sqrt{C_1} \sqrt{R_3} \sqrt{R_L} \sqrt{g_m} \sqrt{C_3 + C_L} \sqrt{R_3 + R_L} \sqrt{C_1 C_3 R_3 R_L + C_1 C_L R_3 R_L}}$
K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.8 LP-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_3 g_m}{C_1 C_L R_1 R_3 s^2 + R_1 g_m + s (C_1 R_1 + C_L R_1 R_3 g_m + C_L R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_3} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_L R_1 R_3 g_m + C_L R_3}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_3}}$
bandwidth: $\frac{C_1 R_1 + C_L R_1 R_3 g_m + C_L R_3}{C_1 C_L R_1 R_3}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.9 LP-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_3 R_L g_m}{C_1 C_L R_1 R_3 R_L s^2 + R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L + s (C_1 R_1 R_3 + C_1 R_1 R_L + C_L R_1 R_3 R_L g_m + C_L R_3 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_3} \sqrt{R_L} \sqrt{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}{C_1 R_1 R_3 + C_1 R_1 R_L + C_L R_1 R_3 R_L g_m + C_L R_3 R_L}$
wo: $\frac{\sqrt{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_3} \sqrt{R_L}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_L + C_L R_1 R_3 R_L g_m + C_L R_3 R_L}{C_1 C_L R_1 R_3 R_L}$
K-LP: $\frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.10 LP-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m}{C_1 C_3 R_1 R_L s^2 + R_1 g_m + s(C_1 R_1 + C_3 R_1 R_L g_m + C_3 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_3 R_1 R_L g_m + C_3 R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_L g_m + C_3 R_L}{C_1 C_3 R_1 R_L}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.11 LP-11 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_L g_m}{R_1 g_m + s^2(C_1 C_3 R_1 R_L + C_1 C_L R_1 R_L) + s(C_1 R_1 + C_3 R_1 R_L g_m + C_3 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_L} \sqrt{C_3 + C_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_3 R_1 R_L g_m + C_3 R_L + C_L R_1 R_L g_m + C_L R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_L} \sqrt{C_3 + C_L}}$
 bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_L g_m + C_3 R_L + C_L R_1 R_L g_m + C_L R_L}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_L} \sqrt{C_3 + C_L} \sqrt{C_1 C_3 R_1 R_L + C_1 C_L R_1 R_L}}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.12 LP-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_3 R_L g_m}{C_1 C_3 R_1 R_3 R_L s^2 + R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L + s(C_1 R_1 R_3 + C_1 R_1 R_L + C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_L} \sqrt{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}{C_1 R_1 R_3 + C_1 R_1 R_L + C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L}$
 wo: $\frac{\sqrt{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_L + C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L}{C_1 C_3 R_1 R_3 R_L}$
 K-LP: $\frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.13 LP-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s^2(C_1 C_3 R_1 R_3 + C_1 C_L R_1 R_3) + s(C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + C_L R_1 R_3 g_m + C_L R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{C_3 + C_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + C_L R_1 R_3 g_m + C_L R_3}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{C_3 + C_L}}$
 bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + C_L R_1 R_3 g_m + C_L R_3}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{C_3 + C_L} \sqrt{C_1 C_3 R_1 R_3 + C_1 C_L R_1 R_3}}$

K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.14 LP-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L + s^2 (C_1 C_3 R_1 R_3 R_L + C_1 C_L R_1 R_3 R_L) + s (C_1 R_1 R_3 + C_1 R_1 R_L + C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L + C_L R_1 R_3 R_L g_m + C_L R_3 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_L} \sqrt{C_3 + C_L} \sqrt{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}{C_1 R_1 R_3 + C_1 R_1 R_L + C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L + C_L R_1 R_3 R_L g_m + C_L R_3 R_L}$
wo: $\frac{\sqrt{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}{\sqrt{C_1 C_3 R_1 R_3 R_L + C_1 C_L R_1 R_3 R_L}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_L + C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L + C_L R_1 R_3 R_L g_m + C_L R_3 R_L}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_L} \sqrt{C_3 + C_L} \sqrt{C_1 C_3 R_1 R_3 R_L + C_1 C_L R_1 R_3 R_L}}$
K-LP: $\frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L R_1 R_3 R_L g_m s + R_1 R_3 g_m}{R_1 g_m + s^2 (C_3 C_L R_1 R_3 R_L g_m + C_3 C_L R_3 R_L) + s (C_3 R_1 R_3 g_m + C_3 R_3 + C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}}{C_3 R_3 + C_L R_3 + C_L R_L}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}}$
bandwidth: $\frac{C_3 R_3 + C_L R_3 + C_L R_L}{C_3 C_L R_3 R_L}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_L R_1 R_3 R_L g_m}{C_3 R_1 R_3 g_m + C_3 R_3 + C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_3 R_1 R_3 R_L g_m s + R_1 R_L g_m}{R_1 g_m + s^2 (C_3 C_L R_1 R_3 R_L g_m + C_3 C_L R_3 R_L) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}}{C_3 R_3 + C_3 R_L + C_L R_L}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}}$
bandwidth: $\frac{C_3 R_3 + C_3 R_L + C_L R_L}{C_3 C_L R_3 R_L}$
K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_3 R_1 R_3 R_L g_m}{C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L + C_L R_1 R_L g_m + C_L R_L}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L R_3 R_L g_m s + R_3 g_m}{g_m + s^2 (C_1 C_L R_3 + C_1 C_L R_L) + s (C_1 + C_L R_3 g_m + C_L R_L g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_L} \sqrt{g_m} \sqrt{R_3 + R_L}}{C_1 + C_L R_3 g_m + C_L R_L g_m} \\ \text{wo: } & \frac{\sqrt{g_m}}{\sqrt{C_1 C_L R_3 + C_1 C_L R_L}} \\ \text{bandwidth: } & \frac{C_1 + C_L R_3 g_m + C_L R_L g_m}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_3 + R_L} \sqrt{C_1 C_L R_3 + C_1 C_L R_L}} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_L R_3 R_L g_m}{C_1 + C_L R_3 g_m + C_L R_L g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_3 R_3 R_L g_m s + R_L g_m}{g_m + s^2 (C_1 C_3 R_3 + C_1 C_3 R_L) + s (C_1 + C_3 R_3 g_m + C_3 R_L g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_3} \sqrt{g_m} \sqrt{R_3 + R_L}}{C_1 + C_3 R_3 g_m + C_3 R_L g_m} \\ \text{wo: } & \frac{\sqrt{g_m}}{\sqrt{C_1 C_3 R_3 + C_1 C_3 R_L}} \\ \text{bandwidth: } & \frac{C_1 + C_3 R_3 g_m + C_3 R_L g_m}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_3 + R_L} \sqrt{C_1 C_3 R_3 + C_1 C_3 R_L}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_3 R_3 R_L g_m}{C_1 + C_3 R_3 g_m + C_3 R_L g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L R_1 R_3 R_L g_m s + R_1 R_3 g_m}{R_1 g_m + s^2 (C_1 C_L R_1 R_3 + C_1 C_L R_1 R_L) + s (C_1 R_1 + C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_3 + R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L} \\ \text{wo: } & \frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_L R_1 R_3 + C_1 C_L R_1 R_L}} \\ \text{bandwidth: } & \frac{C_1 R_1 + C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_3 + R_L} \sqrt{C_1 C_L R_1 R_3 + C_1 C_L R_1 R_L}} \\ \text{K-LP: } & \frac{R_1 R_3 g_m}{R_1 g_m + 1} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_L R_1 R_3 R_L g_m}{C_1 R_1 + C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_3 R_1 R_3 R_L g_m s + R_1 R_L g_m}{R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_L) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3 + R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L} \\ \text{wo: } & \frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_L}} \\ \text{bandwidth: } & \frac{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3 + R_L} \sqrt{C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_L}} \end{aligned}$$

K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_3 R_1 R_3 R_L g_m}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L}$
Qz: None
Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 (C_1 C_L R_1 R_3 g_m + C_1 C_L R_3) + s (C_1 R_1 g_m + C_1 + C_L R_3 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_3} \sqrt{\frac{g_m}{R_1 g_m + 1}}}{C_1 R_1 g_m + C_1 + C_L R_3 g_m}$
wo: $\sqrt{\frac{g_m}{C_1 C_L R_1 R_3 g_m + C_1 C_L R_3}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_1 C_L R_1 R_3 g_m + C_1 C_L R_3}} (C_1 R_1 g_m + C_1 + C_L R_3 g_m)}{\sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_3} \sqrt{\frac{g_m}{R_1 g_m + 1}}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_1 R_1 R_3 g_m}{C_1 R_1 g_m + C_1 + C_L R_3 g_m}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_L g_m s + R_3 R_L g_m}{R_3 g_m + R_L g_m + s^2 (C_1 C_L R_1 R_3 R_L g_m + C_1 C_L R_3 R_L) + s (C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_L R_3 R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_3} \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L}}{C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_L R_3 R_L g_m}$
wo: $\sqrt{\frac{R_3 g_m + R_L g_m}{C_1 C_L R_1 R_3 R_L g_m + C_1 C_L R_3 R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_3 g_m + R_L g_m}{C_1 C_L R_1 R_3 R_L g_m + C_1 C_L R_3 R_L}} (C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_L R_3 R_L g_m)}{\sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_3} \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L}}$
K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_3 R_L g_m}{C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_L R_3 R_L g_m}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_1 R_1 R_L g_m s + R_L g_m}{g_m + s^2 (C_1 C_3 R_1 R_L g_m + C_1 C_3 R_L) + s (C_1 R_1 g_m + C_1 + C_3 R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_L} \sqrt{\frac{g_m}{R_1 g_m + 1}}}{C_1 R_1 g_m + C_1 + C_3 R_L g_m}$
wo: $\sqrt{\frac{g_m}{C_1 C_3 R_1 R_L g_m + C_1 C_3 R_L}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_1 C_3 R_1 R_L g_m + C_1 C_3 R_L}} (C_1 R_1 g_m + C_1 + C_3 R_L g_m)}{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_L} \sqrt{\frac{g_m}{R_1 g_m + 1}}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_1 R_1 R_L g_m}{C_1 R_1 g_m + C_1 + C_3 R_L g_m}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_1 R_1 R_L g_m s + R_L g_m}{g_m + s^2 (C_1 C_3 R_1 R_L g_m + C_1 C_3 R_L + C_1 C_L R_1 R_L g_m + C_1 C_L R_L) + s (C_1 R_1 g_m + C_1 + C_3 R_L g_m + C_L R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_3 \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}}}{C_1 R_1 g_m + C_1 + C_3 R_L g_m + C_L R_L g_m}$

wo: $\sqrt{\frac{g_m}{C_1 C_3 R_1 R_L g_m + C_1 C_3 R_L + C_1 C_L R_1 R_L g_m + C_1 C_L R_L}}$

bandwidth: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_3 \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}}}{\sqrt{C_1} C_3 R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_3 \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L R_1 \sqrt{R_L g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}}}$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_1 R_1 R_L g_m}{C_1 R_1 g_m + C_1 + C_3 R_L g_m + C_L R_L g_m}$

Qz: None

Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_L g_m s + R_3 R_L g_m}{R_3 g_m + R_L g_m + s^2 (C_1 C_3 R_1 R_3 R_L g_m + C_1 C_3 R_3 R_L) + s (C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L}}{C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m}$

wo: $\sqrt{\frac{R_3 g_m + R_L g_m}{C_1 C_3 R_1 R_3 R_L g_m + C_1 C_3 R_3 R_L}}$

bandwidth: $\frac{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L}}{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{R_3 + R_L}}$

K-LP: $\frac{R_3 R_L}{R_3 + R_L}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 R_L g_m}{C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m}$

Qz: None

Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + C_1 C_L R_1 R_3 g_m + C_1 C_L R_3) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_L R_3 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}}}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_L R_3 g_m}$

wo: $\sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + C_1 C_L R_1 R_3 g_m + C_1 C_L R_3}}$

bandwidth: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}}}{\sqrt{C_1} C_3 R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} + \sqrt{C_1} C_L \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}}}$

K-LP: R_3

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 g_m}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_L R_3 g_m}$

Qz: None

Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_L g_m s + R_3 R_L g_m}{R_3 g_m + R_L g_m + s^2 (C_1 C_3 R_1 R_3 R_L g_m + C_1 C_3 R_3 R_L + C_1 C_L R_1 R_3 R_L g_m + C_1 C_L R_3 R_L) + s (C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m + C_L R_3 R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1 C_3 R_1} \sqrt{R_3} \sqrt{R_L} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L} + \sqrt{C_1} C_L R_1 \sqrt{R_3} \sqrt{R_L} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L} + \sqrt{C_1} C_L R_3 R_L \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L}}{C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m + C_L R_3 R_L g_m}$

wo: $\sqrt{\frac{R_3 g_m + R_L g_m}{C_1 C_3 R_1 R_3 R_L g_m + C_1 C_3 R_3 R_L + C_1 C_L R_1 R_3 R_L g_m + C_1 C_L R_3 R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_3 g_m + R_L g_m}{C_1 C_3 R_1 R_3 R_L g_m + C_1 C_3 R_3 R_L + C_1 C_L R_1 R_3 R_L g_m + C_1 C_L R_3 R_L}} (C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m + C_L R_3 R_L g_m)}{\sqrt{C_1 C_3 R_1} \sqrt{R_3} \sqrt{R_L} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L} + \sqrt{C_1} C_L R_1 \sqrt{R_3} \sqrt{R_L} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L} + \sqrt{C_1} C_L R_3 R_L \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L}} \sqrt{R_3 + R_L}}$

K-LP: $\frac{R_3 R_L}{R_3 + R_L}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 R_L g_m}{C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L + C_3 R_3 R_L g_m + C_L R_3 R_L g_m}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, R_3, \infty, \infty, R_L)$

$$H(s) = \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, R_3, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s (C_L R_1 R_3 g_m + C_L R_3) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, R_3, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L + s (C_L R_1 R_3 R_L g_m + C_L R_3 R_L)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, R_3, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L R_1 R_3 R_L g_m s + R_1 R_3 g_m}{R_1 g_m + s (C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m}{R_1 g_m + s (C_3 R_1 R_L g_m + C_3 R_L) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m}{s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_L g_m}{R_1 g_m + s (C_3 R_1 R_L g_m + C_3 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L R_1 R_L g_m s + R_1 g_m}{s^2 (C_3 C_L R_1 R_L g_m + C_3 C_L R_L) + s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L \right)$$

$$H(s) = \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L + s (C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s (C_3 R_1 R_3 g_m + C_3 R_3 + C_L R_1 R_3 g_m + C_L R_3) + 1}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L + s (C_3 R_1 R_3 R_L g_m + C_3 R_3 R_L + C_L R_1 R_3 R_L g_m + C_L R_3 R_L)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$$

$$H(s) = \frac{C_3 R_1 R_3 R_L g_m s + R_1 R_L g_m}{R_1 g_m + s (C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L) + 1}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{s^2 (C_3 C_L R_1 R_3 g_m + C_3 C_L R_3) + s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L R_1 R_3 R_L g_m s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m + C_L R_1 R_L g_m)}{s^2 (C_3 C_L R_1 R_3 g_m + C_3 C_L R_1 R_L g_m + C_3 C_L R_3 + C_3 C_L R_L) + s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \infty, \infty, R_L \right)$$

$$H(s) = \frac{R_3 R_L g_m}{R_3 g_m + R_L g_m + s (C_1 R_3 + C_1 R_L)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m}{s^2 (C_1 C_3 + C_1 C_L) + s (C_3 g_m + C_L g_m)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L R_L g_m s + g_m}{C_1 C_3 C_L R_L s^3 + s^2 (C_1 C_3 + C_1 C_L + C_3 C_L R_L g_m) + s (C_3 g_m + C_L g_m)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L R_3 R_L g_m s + R_3 g_m}{C_1 C_3 C_L R_3 R_L s^3 + g_m + s^2 (C_1 C_3 R_3 + C_1 C_L R_3 + C_1 C_L R_L + C_3 C_L R_3 R_L g_m) + s (C_1 + C_3 R_3 g_m + C_L R_3 g_m + C_L R_L g_m)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 R_3 g_m s + g_m}{C_1 C_3 C_L R_3 s^3 + s^2 (C_1 C_3 + C_1 C_L + C_3 C_L R_3 g_m) + s (C_3 g_m + C_L g_m)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 R_3 R_L g_m s + R_L g_m}{C_1 C_3 C_L R_3 R_L s^3 + g_m + s^2 (C_1 C_3 R_3 + C_1 C_3 R_L + C_1 C_L R_L + C_3 C_L R_3 R_L g_m) + s (C_1 + C_3 R_3 g_m + C_3 R_L g_m + C_L R_L g_m)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L R_3 R_L g_m s^2 + g_m + s (C_3 R_3 g_m + C_L R_L g_m)}{s^3 (C_1 C_3 C_L R_3 + C_1 C_3 C_L R_L) + s^2 (C_1 C_3 + C_1 C_L + C_3 C_L R_3 g_m + C_3 C_L R_L g_m) + s (C_3 g_m + C_L g_m)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, \infty, R_L \right)$$

$$H(s) = \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L + s (C_1 R_1 R_3 + C_1 R_1 R_L)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m}{s^2 (C_1 C_3 R_1 + C_1 C_L R_1) + s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L R_1 R_L g_m s + R_1 g_m}{C_1 C_3 C_L R_1 R_L s^3 + s^2 (C_1 C_3 R_1 + C_1 C_L R_1 + C_3 C_L R_1 R_L g_m + C_3 C_L R_L) + s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L R_1 R_3 R_L g_m s + R_1 R_3 g_m}{C_1 C_3 C_L R_1 R_3 R_L s^3 + R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + C_1 C_L R_1 R_3 + C_1 C_L R_1 R_L + C_3 C_L R_1 R_3 R_L g_m + C_3 C_L R_3 R_L) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + C_L R_1 R_3 g_m + C_L R_1 R_L g_m + C_L R_3 + C_L R_L) + 1}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{C_1 C_3 C_L R_1 R_3 s^3 + s^2 (C_1 C_3 R_1 + C_1 C_L R_1 + C_3 C_L R_1 R_3 g_m + C_3 C_L R_3) + s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_3 R_1 R_3 R_L g_m s + R_1 R_L g_m}{C_1 C_3 C_L R_1 R_3 R_L s^3 + R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_L + C_1 C_L R_1 R_L + C_3 C_L R_1 R_3 R_L g_m + C_3 C_L R_3 R_L) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_1 R_L g_m + C_3 R_3 + C_3 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_3 C_L R_1 R_3 R_L g_m s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m + C_L R_1 R_L g_m)}{s^3 (C_1 C_3 C_L R_1 R_3 + C_1 C_3 C_L R_1 R_L) + s^2 (C_1 C_3 R_1 + C_1 C_L R_1 + C_3 C_L R_1 R_3 g_m + C_3 C_L R_1 R_L g_m + C_3 C_L R_3 + C_3 C_L R_L) + s (C_3 R_1 g_m + C_3 + C_L R_1 g_m + C_L)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_L g_m s + R_3 R_L g_m}{R_3 g_m + R_L g_m + s (C_1 R_1 R_3 g_m + C_1 R_1 R_L g_m + C_1 R_3 + C_1 R_L)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 R_1 g_m s + g_m}{s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + C_1 C_L R_1 g_m + C_1 C_L) + s (C_3 g_m + C_L g_m)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_L R_1 R_L g_m s^2 + g_m + s (C_1 R_1 g_m + C_L R_L g_m)}{s^3 (C_1 C_3 C_L R_1 R_L g_m + C_1 C_3 C_L R_L) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + C_1 C_L R_1 g_m + C_1 C_L + C_3 C_L R_L g_m) + s (C_3 g_m + C_L g_m)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_L R_1 R_3 R_L g_m s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m + C_L R_3 R_L g_m)}{g_m + s^3 (C_1 C_3 C_L R_1 R_3 R_L g_m + C_1 C_3 C_L R_3 R_L) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + C_1 C_L R_1 R_3 g_m + C_1 C_L R_1 R_L g_m + C_1 C_L R_3 + C_1 C_L R_L + C_3 C_L R_3 R_L g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_L R_3 g_m + C_L R_L g_m)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 g_m s^2 + g_m + s (C_1 R_1 g_m + C_3 R_3 g_m)}{s^3 (C_1 C_3 C_L R_1 R_3 g_m + C_1 C_3 C_L R_3) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + C_1 C_L R_1 g_m + C_1 C_L + C_3 C_L R_3 g_m) + s (C_3 g_m + C_L g_m)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 R_L g_m s^2 + R_L g_m + s (C_1 R_1 R_L g_m + C_3 R_3 R_L g_m)}{g_m + s^3 (C_1 C_3 C_L R_1 R_3 R_L g_m + C_1 C_3 C_L R_3 R_L) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_L g_m + C_1 C_3 R_3 + C_1 C_3 R_L + C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_3 C_L R_3 R_L g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 R_L g_m + C_L R_L g_m)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_3 C_L R_1 R_3 R_L g_m s^3 + g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_L R_1 R_L g_m + C_3 C_L R_3 R_L g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + C_L R_L g_m)}{s^3 (C_1 C_3 C_L R_1 R_3 g_m + C_1 C_3 C_L R_1 R_L g_m + C_1 C_3 C_L R_3 + C_1 C_3 C_L R_L) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + C_1 C_L R_1 g_m + C_1 C_L + C_3 C_L R_3 g_m + C_3 C_L R_L g_m) + s (C_3 g_m + C_L g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_L R_1 R_3 R_L g_m s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m + C_L R_3 R_L g_m)}{g_m + s^2 (C_1 C_L R_1 R_3 g_m + C_1 C_L R_1 R_L g_m + C_1 C_L R_3 + C_1 C_L R_L) + s (C_1 R_1 g_m + C_1 + C_L R_3 g_m + C_L R_L g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_L} R_1 R_3 g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_L} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_L} R_3 \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_L} R_L \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}}{C_1 R_1 g_m + C_1 + C_L R_3 g_m + C_L R_L g_m} \\ \text{wo: } & \sqrt{\frac{g_m}{C_1 C_L R_1 R_3 g_m + C_1 C_L R_1 R_L g_m + C_1 C_L R_3 + C_1 C_L R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_m}{C_1 C_L R_1 R_3 g_m + C_1 C_L R_1 R_L g_m + C_1 C_L R_3 + C_1 C_L R_L}} (C_1 R_1 g_m + C_1 + C_L R_3 g_m + C_L R_L g_m)}{\sqrt{C_1} \sqrt{C_L} R_1 R_3 g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_L} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_L} R_3 \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_L} R_L \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L} \\ \text{K-BP: } & \frac{C_1 R_1 R_3 g_m + C_L R_3 R_L g_m}{C_1 R_1 g_m + C_1 + C_L R_3 g_m + C_L R_L g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L}} \end{aligned}$$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 R_L g_m s^2 + R_L g_m + s (C_1 R_1 R_L g_m + C_3 R_3 R_L g_m)}{g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_L g_m + C_1 C_3 R_3 + C_1 C_3 R_L) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 R_L g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_3} R_1 R_3 g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_3} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_3} R_3 \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_3} R_L \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 R_L g_m} \\ \text{wo: } & \sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_L g_m + C_1 C_3 R_3 + C_1 C_3 R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_L g_m + C_1 C_3 R_3 + C_1 C_3 R_L}} (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 R_L g_m)}{\sqrt{C_1} \sqrt{C_3} R_1 R_3 g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_3} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_3} R_3 \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}} + \sqrt{C_1} \sqrt{C_3} R_L \sqrt{\frac{g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L}}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & \frac{R_1 R_3 R_L g_m}{R_1 R_3 g_m + R_1 R_L g_m + R_3 + R_L} \\ \text{K-BP: } & \frac{C_1 R_1 R_L g_m + C_3 R_3 R_L g_m}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 R_L g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3}} \end{aligned}$$

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